BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE **TITLE:** STANDARD CHEMICAL TESTING PROCEDURE FOR BIOMEDICAL CO-TRIMOXAZOLE SUSPENSION (50ml)		
DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/021**	**PAGE NO.:**	**Page 1 of 6**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
REVIEW DATE:	**04/01/2026**	**SUPERSEDE:**	**00**

S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Sulphamethoxazole** **Label Claim: 200mg/5ml**	**METHOD 1: Sulphamethoxazole** Reagents: a. P – Dimethylaminobenzaldehyde solution: 200mg of p – methylaminobenzaldehyde is dissolved in 25ml of acetone or methanol, 5ml of 1N hydrochloric acid is added and then diluted to 100ml with distilled water. b. 25%v/v solution of acetone or methanol in water. ***Standard solution***: Weigh equivalent of 100mcg/ml of Sulphamethoxazole in aqueous acetone or methanol ***Sample solution:*** Accurately weigh, a quantity equivalent ot 200mg of Sulphamethoxazole, wash the pipette with two 5ml portions of water and mix with the main solution. Add 25ml of acetone, shake and dilute to 100ml with water. 5ml of the resultant solution is further diluted with aqueous acetone or methanol to 100ml. (100mcg/ml)

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	**QUALITY CONTROL SENIOR ANALYST**		
REVIEWED BY:	**QUALITY CONTROL HEAD**		
APPROVED BY:	**QUALITY ASSURANCE MANAGER**		

Procedure:

To 5ml of sample and standard solutions, add 10ml of the reagent and make up the volume to 25ml with acetone. Allow to stand for 40 minutes at room temperature

Measure the absorbance of both the solutions at 450nm against reagent blank.

% Assay of Sulphamethoxazole = $\frac{absorbance of sample}{absorbance of standard}$ x 100

METHOD 2:

Standard solution:

Weigh equivalent of 100mg/ml of Sulphamethoxazole in 1N sodium hydroxide. Further dilutions are done with water to get 10mcg/ml.

Sample solution:

Collect quantitatively alkaline layer retained during essay to trimethoprim and dilute to 100ml with 1N sodium hydroxide. Further dilutions are done with water to get a concentration of 10mcg/ml.

Measure the extinction of both the sample and standard solutions at maxima at about 260nm and calculate the results by comparison.

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

		% Assay of Sulphamethoxazole = $\frac{absorbance of sample}{absorbance of standard}$ x 100
3.	**Trimethoprim** **Label Claim: 40mg/5ml**	**METHOD 1:** ***Standard solution:*** Weigh 40mg of Trimethoprim into a separating funnel ***Sample solution:*** Transfer accurately weighed portion of the sample equivalent to 40mg of trimethoprim to a separating funnel. In case of suspensions, shake the sample well for 10 minutes before withdrawing the sample for analysis. Add 30ml of 1N sodium hydroxide and shake for 1 min. Extract the aqueous layer with 25ml chloroform thrice. (Keep aqueous for Sulphamethoxazole, if method of sulphamethoxazole is required). Wash with 10ml of 0.1N sodium hydroxide. Extract the chloroform layer with 50ml of 1M acetic acid x 4. Wash with 10ml chloroform x 2 Make up to 250ml with 1M Acetic Acid, take 10ml and make up to 100ml with 1M Acetic Acid. Measure absorbance at 271nm **% Assay of Trimethoprim** = $\frac{absorbance\ of\ sample}{absorbance\ of\ standard}$ x 100 ***REFERENCE:***

		Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 2
4.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample. - Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an "S" or "8" shape. - Allow the media to completely solidify. - Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days. **<u>TEST FOR ABSENCE OF PATHOGEN</u>** - Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth - Transfer 10ml into 90ml of Tryptone soya broth - Incubate at 30-35^{0}c for 18-24hours

- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.
- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test.
- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\textbf{CFU/mL} = \frac{\textbf{Total number of colonies obtained x dilution factor}}{\textbf{Volume of sample used (aliquot)}}$$

		- This will give the total number of viable microbial cells present in the given sample. - For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated. - If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats. - The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105** - If the colonies are fused or the whole plate is covered with a single colony, then report as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

CURRENT REV	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		